

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) endeavour to elucidate how infinitesimal perturbations of the input suffice to overturn a model’s prediction; yet a persistent shortcoming of prevailing methods is their propensity to modify substantially more features than is strictly necessary, thereby attenuating both the interpretive clarity and the practical actionability of the resulting explanation. To redress this deficiency, we advance *COLA*, a post-hoc framework that is agnostic to both the underlying model and the CE generator and that refines any pre-existing counterfactual by first computing, via optimal transport (OT), a coupling between the factual and counterfactual sets, and subsequently exploiting this coupling to steer a Shapley-based attribution (p -SHAP) whose office is to isolate a parsimonious set of edits without forfeiting the targeted effect. On the theoretical side, we establish that OT minimises an upper bound on the W_1 divergence between factual and counterfactual outcomes, and, moreover, that under mild regularity conditions the refined counterfactuals are provably guaranteed never to drift farther from the factuals than their unrefined counterparts. On the empirical side, across a suite comprising four datasets, twelve models, and five CE generators, *COLA* attains the identical target effects while expending merely 26–45% of the feature edits demanded by the originals; and on a small-scale benchmark it demonstrably approaches optimality.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) constitutes an indispensable prerequisite for rendering artificial intelligence systems transparent and, by extension, trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this broad enterprise, feature attributions (FA) methods—of which Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) are the canonical exemplar—serve to quantify the extent to which each input feature contributes to the output of a machine learning (ML) model, thereby affording a principled means of distilling otherwise opaque models by foregrounding their most influential covariates. In the clinical setting, for instance, Shapley values can surface determinative factors such as age and medical history, thus assisting practitioners in apprehending the rationale behind a model’s decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). A complementary paradigm, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), instead illuminates how modest perturbations of the input features may precipitate divergent outcomes. Although the literature to date has furnished hundreds of CE algorithms (Guidotti, 2022; Verma et al., 2020), the aspiration of identifying a single algorithm that would accommodate every use case remains largely chimerical, precisely because each such method is calibrated to its own idiosyncratic constellation of scenarios, objectives, and tasks. The operative objective may, in one formulation, amount to procuring a single counterfactual instance for each factual instance, whereas in another it may consist in treating multiple instances collectively as a group and seeking one or several counterfactual instances per factual observation. In certain cases the locus of concern is the dataset in its entirety, the aim being to identify global CE that prescribe the direction along which the factual instances ought to be displaced so as to elicit the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024); in yet other cases the ensemble of factual instances is construed as a distribution, whereupon the objective becomes that of recovering a counterfactual distribution that preserves the morphology of the factual distribution (You et al., 2025) while incurring commensurate costs. Beyond these distinctions of objective, some CE algorithms

presuppose differentiable models, while others are contrived expressly for tree-based or ensemble architectures.

Problem Description and Challenges Although FA and CE are alike indispensable to the interpretability and accountability of AI models, an exclusive reliance upon either one entails characteristic infirmities: FA taken in isolation seldom furnishes the tangible, actionable steps that constitute a satisfactory explanation, whereas CE taken in isolation is apt to incorporate superfluous—and hence impractical—feature changes. Motivated by this complementarity, the present paper undertakes a general and comprehensive problem that builds upon the extensive foundations laid down in the literature (see Appendix A). More precisely, we seek to adjudicate the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{e}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{e}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{e}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{R}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to augment its volume of user registrations, having amassed data on user interactions—among them the monetary expenditure ($\textcircled{e}$), the number of clicks ($\blacktriangleright$), and the registration status ($\textcircled{R}$). In the original data ($\mathbf{x}$), no user is registered. The action plans $\mathbf{z}'$ and $\mathbf{z}''$ each recalibrate the user characteristics so as to realise the desired outcome ($\mathbf{y}^*$) of universal registration. Although both plans procure an identical, one-half counterfactual effect, $\mathbf{z}''$ accomplishes it through demonstrably fewer modifications than $\mathbf{z}'$.

Three principal challenges continue to encumber any attempt to resolve this problem. First, it would be unrealistic to expect a solitary CE algorithm to satisfy every requirement universally, inasmuch as the problem is inherently task-specific. Second, the approach ought not to predicate itself upon stringent assumptions concerning the model—such as the requirement of differentiability or of some distinguished structure—if its applicability across a broad spectrum of models is to be preserved. Third, FA of the feature-importance variety is liable to prove misleading, owing to the want of coherence between the FA scores and the modifications that actually engender the counterfactual effect; stated otherwise, it is ineffective to perform FA independently of CE in order to select the most consequential features to alter. We shall subsequently demonstrate—in Result II of Section 6—that such decoupling can precipitate counterproductive feature modifications (equivalently, actions on the part of users or stakeholders), for the features adjudged important in the general sense need not coincide with the specific pathways along which the desired counterfactual outcomes are attained.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE across heterogeneous models and CE generators by enlisting an optimal transport (OT)–induced coupling between the factual and counterfactual sets to marshal the Shapley attributions. This construction gives rise to p -SHAP, which subsumes several commonly employed Shapley methods as special cases obtained under appropriate couplings and which, in addition, exposes a modular interface for attribution and edit selection. Our contributions are furthermore underwritten by theoretical guarantees: OT minimises an upper bound on the W_1 divergence, and, under mild conditions, the refined counterfactuals are provably no farther removed from the factuals than the originals. Empirically, across four datasets, twelve models, and five CE methods, COLA equipped with p -SHAP realises the same target effects while expending only 26–45% of the original feature edits; and on a small-scale benchmark it proves near-optimal.

To the best of our knowledge, this constitutes the first method advanced for systematically addressing the problem depicted in Figure 1 *without committing to any particular CE algorithm or model*.

2 PROBLEM FORMULATION

We now render precise the problem adumbrated in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, let $\mathbf{x}$ denote any observed (factual) data comprising n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$), and let $\mathbf{y}^*$ denote the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimisation consists in the pursuit of a (group of) counterfactual data instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$)—subject to a ceiling of C on the number of admissible feature changes—such that the attendant model output(s) approximate $\mathbf{y}^*$ as closely as may be. Writing D for a divergence function that gauges the dissimilarity between two entities, the problem admits the following formulation.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) in concert with the constraint equation (1b) articulates the canonical CE optimisation: namely, $\mathbf{z}$ is required to drive $f(\mathbf{z})$ towards $\mathbf{y}^*$ while nonetheless remaining proximate to $\mathbf{x}$, so that $\mathbf{z}$ may thereafter serve as a contrastive reference elucidating why $f(\mathbf{x})$ fails to attain $\mathbf{y}^*$. We deliberately refrain from restricting the function D to any particular class of divergence, thereby preserving its generality; instantiations of D may accordingly include the Euclidean distance, OT, maximum mean discrepancy (MMD), or differences of the model outcome in mean or median. The remaining conditions equation (1c)–equation (1e) then constitute the action-constrained CE optimisation. Superimposed upon the counterfactual data $\mathbf{z}$, we additionally optimise an indicator variable $\mathbf{c}$, contrived such that z_{ik} is precluded from changing if and only if $c_{ik} = 0$, with at most C changes sanctioned by equation (1c). On inspection of equation (1d) and equation (1e), one observes that whenever $c_{ik} = 0$ the equality $x_{ik} = z_{ik}$ obtains and no modification occurs at position (i, k) ; whenever $c_{ik} = 1$, by contrast, the constant M —taken sufficiently large—confers upon z_{ik} ample latitude to vary.

To solve equation (1), we have recourse to FA as the instrument for identifying the most influential features and thereby recovering the modification indicator variable $\mathbf{c}$. The section that follows introduces the commonly employed Shapley value methods for FA, which, in conjunction with the method we later propose, are incorporated into our algorithmic framework COLA in order to yield the refined counterfactual $\mathbf{z}$. The problem, it bears emphasising, is computationally recalcitrant even in the degenerate case $d = 1$ for linear models; see Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section rehearses the commonly employed Shapley value methods for FA which, together with the method we later propose, are woven into our algorithmic framework COLA. We begin by expounding the notion of the Shapley value as it originates in game theory, and proceed thereafter to a discussion of the various Shapley methods customarily deployed for FA; the interested reader may consult (Sundararajan & Najmi, 2020) for a comprehensive survey of these variants.

In cooperative game theory, a coalitional game is characterised by a finite set of players (in our context, features), denoted $\mathcal{F} = \{1, 2, \dots, d\}$, together with a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, which represents the aggregate payoff attainable by the members of S through cooperation, subject to the normalisation $v(\emptyset) = 0$. The Shapley value, defined below in terms of v , is the foundational construct of this framework, furnishing an equitable prescription for apportioning the overall payoff $v(\mathcal{F})$ of the grand coalition among the individual players.

Antecedent to defining the Shapley value, let $\Delta(k, S)$ denote the incremental value that a player (feature) k confers upon a coalition S —a quantity coextensive with the notion of marginal contribution—defined as the additional payoff accruing to the coalition by virtue of the inclusion of player k ; that is,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k is obtained by averaging the normalised marginal contributions $\Delta(k, S)$ of feature k over all subsets S of the feature set that exclude k , the normalisation factor being the reciprocal of the binomial coefficient $\binom{d-1}{|S|}$, in which d is the total number of features. By summing these contributions across all subsets and dividing by d , the expression effects a fair and axiomatically grounded distribution of payoffs among the players, each feature’s allotment faithfully reflecting its contribution relative to the entire collection of features.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for the portion comprising only the features in S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for the complementary remainder, and the same convention governs the analogous notations throughout. We now introduce the commonly employed Shapley value methods.

Baseline Shapley (B-SHAP). This approach predicates itself upon a designated baseline data point $\mathbf{r}_j$, which serves as the reference for any $\mathbf{x}_i$. The set function is defined to be

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

wherein the absence of a feature is modelled by its value in the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method presupposes an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This approach, a variant of B-SHAP, is implicitly invoked in (Lundberg & Lee, 2017) as well as in the widely used “SHAP” library, and is governed by the following equation.

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ denotes the background distribution, which is conventionally taken to be the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This constitutes an extension of RB-SHAP obtained by identifying the distribution $\mathcal{D}$ with the counterfactual distribution conditioned upon each individual factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method, adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and there shown to confer advantages for contrastive FA, presupposes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalise equation (4) through equation (6) by incorporating an algorithm A_{Prob} that returns their joint probability.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, and in contradistinction to prior methods that repose upon random baselines, p -SHAP resolves the alignment problem by identifying the optimal coupling via OT—a manoeuvre that recasts feature attribution not as the mere decomposition of a prediction but as a transport problem whose solution minimises the very cost of explanation.

¹In the first place, p -SHAP degenerates to B-SHAP in equation (4) whenever A_{Prob} specifies a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that encodes an $i \leftrightarrow j$ alignment for any $\mathbf{x}_i, \mathbf{r}_j$. In the second place, p -SHAP degenerates to RB-SHAP in equation (5) whenever A_{Prob} is defined independently of CE yet associated with an arbitrary distribution $\mathcal{D}$. And in the third place, p -SHAP degenerates to CF-SHAP in equation (6) whenever A_{Prob} is predicated upon a known distribution of CE.

It is to be remarked that A_{prob} is independent of the CE algorithm and depends solely upon the generated counterfactual $\mathbf{r}$. By confining its attention exclusively to these fixed components, p -SHAP secures consistency in FA without being perturbed by the variability inherent in disparate CE generation processes—a property that marks its principal point of departure from CF-SHAP. Contrary to prevailing expectation, we demonstrate that OT can prove more efficacious than a reliance upon a counterfactual distribution defined by a CE generation mechanism, as pursued by [Albini et al. \(2022\)](#), in Result II of Section 6 below.

In particular, one of the focal concerns of this paper is the OT problem (equivalently, the 2-Wasserstein divergence) defined below, the transportation plan $\mathbf{p}_{\text{OT}}$ resulting from its solution being adopted as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ within p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$ respectively, while $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ designates the set of joint distributions (that is, the totality of admissible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ furnishes the entropic regularisation, with $\varepsilon \geq 0$ serving as its coefficient; the presence of such regularisation ($\varepsilon > 0$) materially expedites the computation of OT.

Theoretical Aspects of p -SHAP OT minimises the aggregate feature-modification cost (that is, the cost of transporting $\mathbf{x}$ towards $\mathbf{r}$) under the alignment it induces between the factual $\mathbf{x}$ and the counterfactual $\mathbf{r}$, a desideratum that corresponds precisely to our objective of procuring feature modifications that realise the counterfactual outcomes at minimal expense. This correspondence may be fortified theoretically under the assumption that the predictive model f is Lipschitz continuous. In Theorem 4.1 below (with proof deferred to Appendix C), we establish that the transportation plan $\mathbf{p}_{\text{OT}}$ employed in p -SHAP is efficacious in minimising an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded above by the Lipschitz constant (under the presumption that f is Lipschitz continuous) multiplied by the square root of the minimised expected cost of transporting $\mathbf{x}$ towards $\mathbf{r}$ —namely $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, wherein the quantities p_{ij} ($j = 1, 2, \dots, m$) encode the degree to which the feature values of $\mathbf{x}_i$ are to be adjusted towards those of one or several $\mathbf{r}_j$. In practical terms, this signifies that within p -SHAP the OT plan $\mathbf{p}_{\text{OT}}$ supplies a strategy for adjusting the feature values of $\mathbf{x}$ towards those of $\mathbf{r}$ in such a manner as to minimise the expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. By comparison with any alternative modification plan ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ incurs the least attainable cost, which in turn furnishes the tightest upper bound—proportional to this cost—on the violation of the counterfactual effect $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (*p -SHAP Towards Counterfactual Effect*). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds only the transport cost, this bound operates as a convex surrogate for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem; by concentrating the attribution mass upon the most efficient paths, the OT coupling effectively steers the greedy selection in Algorithm 1 towards discarding the non-essential features. Beyond this, p -SHAP is conceptually sound in ascribing the causal behaviour to the modifications of the characteristics, as asserted in Theorem 4.2 below (with proof in Appendix D).

Theorem 4.2 (*Interventional Effect of p -SHAP*). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the*

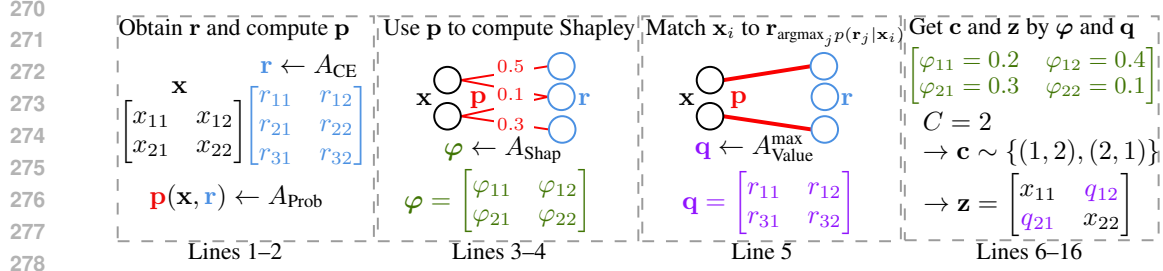


Figure 2: [An illustration of COLA] The figure depicts the manner in which COLA arrives at $\mathbf{c}$ and $\mathbf{z}$ for equation (1). For the sake of illustration we adopt $A_{\text{Value}}^{\max}$ in line 5, on account of its simplicity. In lines 6–16 we posit $C = 2$, whereupon the sampling yields exactly two positions for modification in accordance with the probability matrix φ .

intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

We further remark that p -SHAP inherits the desirable axioms of B-SHAP and RB-SHAP, a circumstance that renders it an effective instrument for feature attribution. The corresponding proof is omitted, and we refer the reader to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the axiomatic characterisation of the Shapley value.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, set out in Algorithm 1 below, is directed at solving equation (1) and is erected upon four categories of algorithms. We expound Algorithm 1 in detail below, in conjunction with the illustration provided in Figure 2.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$
1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
9: **for** $i \leftarrow 1$ to n **do**
10: **for** $k \leftarrow 1$ to d **do**
11: **if** $c_{ik} = 1$ **then**
12: $z_{ik} \leftarrow q_{ik}$
13: **end if**
14: **end for**
15: **end for**
16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} receives as input the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ , and returns a counterfactual $\mathbf{r}$ that remains proximate to $\mathbf{x}$, satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (*Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). This task is discharged by an algorithm A_{Prob} , which, taking $\mathbf{x}$ and $\mathbf{r}$ as input, emits a matrix representing the joint distribution over the n and m data points of $\mathbf{x}$ and $\mathbf{r}$ respectively. The joint distribution $\mathbf{p}$ encodes an alignment relationship (or matching) between the factual and counterfactual rows, and it is subsequently enlisted in Line 5 to compute the values from which $\mathbf{z}$ will ultimately be composed. As adumbrated in Section 3, A_{Prob} may be grounded in OT so as to compute a joint distribution minimising the OT distance between $\mathbf{x}$ and $\mathbf{r}$ whenever the alignment between their rows is unknown; otherwise, it is advisable to select a joint distribution that faithfully mirrors the alignment between the rows of $\mathbf{x}$ and $\mathbf{r}$.

Lines 3–4 (*p-SHAP FA*). We invoke equation (7) to compute the Shapley value for $\mathbf{x}$. The joint distribution $\mathbf{p}$ may here be enlisted (though it is not obligatory to do so) in order to align each row of $\mathbf{x}$ appropriately with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ furnish the most representative contrastive reference for the pertinent row of $\mathbf{x}$; numerically, this alignment exerts a pronounced influence upon our contrastive FA. The Shapley values (arranged as a matrix) of $\mathbf{x}$ are thereupon taken elementwise in absolute value and normalised so as to sum to unity, the resulting matrix φ constituting our FA.

Line 5 (*Computing feature values*). This task devolves upon the algorithm A_{Value} , which takes as input the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$. For any row i of $\mathbf{x}$, A_{Value} selects one or several row(s) of $\mathbf{r}$ to serve as references for effecting changes in $\mathbf{x}$, and returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ each of whose elements q_{ik} constitutes a counterfactual candidate for x_{ik} ($\forall i, k$). We introduce below the two instantiations $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, corresponding respectively to the selection of a single row and of multiple rows.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the row of $\mathbf{r}$ carrying the highest probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a convex combination—that is, a weighted average—of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Recall that the non-negative matrix φ has been normalised to sum to unity; it may consequently be construed as a policy for selecting the positions in $\mathbf{x}$ at which value replacement is to occur (that is, $c_{ik} = 1$), precisely as line 7 prescribes. For every i and k with $c_{ik} = 1$, the entry x_{ik} is thereupon modified to q_{ik} , and the resulting matrix is returned as $\mathbf{z}$ which, together with $\mathbf{c}$, constitutes the optimised solution of the problem in equation (1).

By virtue of Theorem 4.1, COLA is contrived so as to minimise the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ in accordance with the feature-attribution results, which single out the most consequential features to adjust in order to realise the desired counterfactual effect. The theorem below (with proof in Appendix E) establishes that the refined $\mathbf{z}$ produced by the COLA framework honours the constraint equation (1b) in the typical scenario $n = m$, the Frobenius norm serving as the distance measure. Empirical corroboration of the broader applicability of this conclusion is furnished in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the algorithmic complexity of A_{CE} . For the algorithm A_{Shap} , suppose that weighted linear regression is employed to estimate the Shapley values, and let M_{Shap} denote the number of sampled subsets. The complexity of COLA as a function of n, m, d , and the regularisation parameter ε of entropic OT is then $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ should $A_{\text{Value}}^{\max}$ be employed and $N = \mathcal{O}(nmd)$ should $A_{\text{Value}}^{\text{avg}}$ be employed. The reader is referred to Appendix F for the attendant details.

6 NUMERICAL RESULTS

This section appraises the efficacy of COLA in addressing the problem posed in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual furnished by a CE method A_{CE} . We enlist four distinct divergence functions: OT quantifies the distance between entire distributions; MMD quantifies the divergence between the means of two distributions embedded in a high-dimensional feature space; and the absolute mean difference (MeanD) and absolute median difference (MedianD) quantify the divergence in mean and median respectively. The numerical results are marshalled to substantiate the following: I) *the efficacy of COLA with respect to modification minimality*; II) *the superiority of p -SHAP over competing Shapley methods with respect to modification minimality*; III) *the near-optimal performance of COLA*.

Experiment Setup The experiments are conducted over 4 datasets pertaining to binary classification tasks, 5 CE algorithms devised for divergent goals, and 12 classifiers, as enumerated in Table 1, wherein a combination of a dataset, an A_{CE} algorithm, and a model constitutes an “experiment scenario”.

Table 1: [Experiment Scenarios Setup] Four datasets serve to benchmark five CE algorithms across 12 models. A “scenario” is defined as a combination of a dataset, an A_{CE} algorithm, and a model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table delineates the 6 methods placed in comparison, colour-coded so as to accord with Figure 3. Each method is instantiated within an experiment scenario, as defined in Table 1, for the purpose of benchmarking.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We introduce, in brief, the several A_{CE} appearing in Table 1. DiCE and KNN are data-instance-based CE methods, each yielding counterfactual(s) for every factual instance; AReS and GlobeCE are group-based CE methods, which procure a collection of counterfactual instances for the factual data taken collectively as a group; and Discount construes the factual instances as an empirical distribution and seeks a counterfactual distribution residing in its proximity.

Table 2 delineates 6 methods, of which CF- p_{OT} is the proposed p -SHAP and the remainder serve as baselines. Each is instantiated across numerous experiment scenarios drawn from Table 1 and thereby

benchmarked comprehensively, being determined by a particular combination of the three algorithms A_{Prob} , A_{Value} , and A_{Shap} . By way of illustration, the first row RB- p_{Uni} adopts the uniform distribution as the algorithm A_{Prob} for computing $\mathbf{p}$, whereupon the computed $\mathbf{p}$ is fed in turn into equation (9) or equation (10) of A_{Value} for the computation of $\mathbf{q}$, while A_{Shap} is taken to be RB-SHAP.

These methods have been assembled with deliberate care so as to support the ablation studies. In the first place, RB- p_{Uni} differs from CF- p_{Uni} in that the latter avails itself of CE information whereas the former does not. In the second place, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all employ CE within FA, albeit with differing joint distributions, and we demonstrate below that the distribution computed by OT outperforms the alternatives by a substantial margin. In the third place, in order to ascertain that OT proves useful because it apprises A_{Shap} of the factual-counterfactual alignment, and not on account of any extraneous factor, we juxtapose CF- p_{OT} —which exploits such an alignment—against RB- p_{OT} , which does not. Finally, CF- p_{Ect} represents the special case in which each counterfactual issues from a known source, this being enlisted as an exact factual-counterfactual alignment and thereby rendering CF-SHAP coincident with B-SHAP.

Table 3: [COLA for Modification Minimality] The table reports the number of modified features in $\mathbf{z}$ engendered by each method at the point where $f(\mathbf{z})$ attains 80% of the counterfactual effect of $f(\mathbf{r})$. The result for each method is obtained by averaging over 4 randomly selected scenarios from Table 1, with 100 runs per scenario. The symbol “-” signifies that the target counterfactual effect could not be achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I: COLA achieves substantial action reduction at the price of only a marginal loss in counterfactual effect In Table 3 we impose upon $\mathbf{z}$ the requirement that $f(\mathbf{z})$ attain 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ². It is to be observed that COLA discharges this requirement effectively, exacting appreciably fewer actions in $\mathbf{z}$ (that is, feature modifications in $\mathbf{z}$) than the original CE $\mathbf{r}$. Under COLA one may anticipate recourse to merely 13%–25% of the feature changes (measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$) in order to secure a counterfactual effect of 80%. In particular, COLA equipped with p -SHAP (namely CF- p_{OT}) alone attains the objective of a 100% counterfactual effect, and does so with only 26%–45% of the feature changes present in the original $\mathbf{r}$. p -SHAP, above all, engenders the most pronounced action minimality, a matter analysed in detail below.

²Observe that, by definition, $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect inasmuch as $\mathbf{y}^* = f(\mathbf{r})$. To render precise a counterfactual effect of 80%, one considers the proportion of divergence expunged by the refined CE $\mathbf{z}$; that is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D taken to be OT.

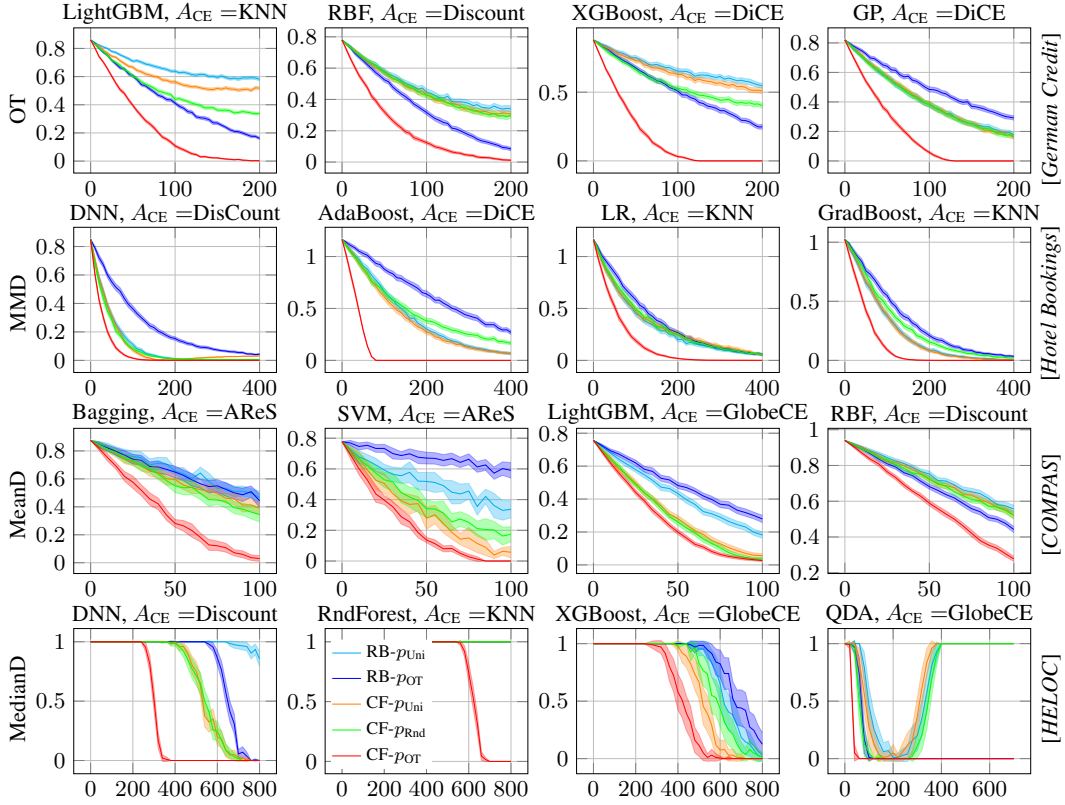


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C . Each experiment comprises 100 runs, and the shaded regions delineate the 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is employed for *HELOC* and *COMPAS*, whereas $A_{\text{Value}}^{\text{max}}$ is employed for *German Credit* and *Hotel Bookings*. The legend contained within “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” pertains to all plots.

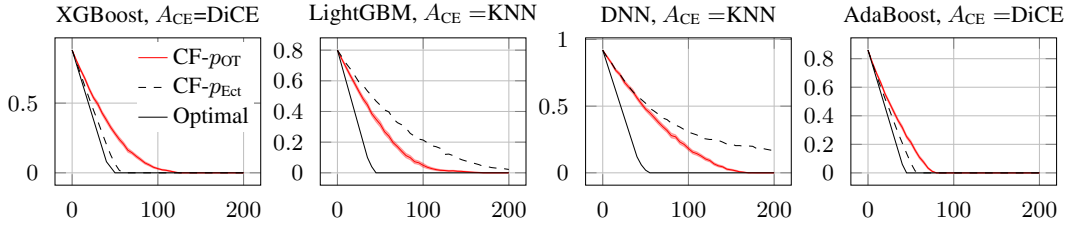


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C , with D taken to be *MeanD*.

Result II: p -SHAP outperforms the competing Shapley methods in the realisation of the counterfactual effect We present in Figure 3 the evaluation of the several Shapley methods of equation (4)–equation (7), the abscissa recording the number of allowed feature changes C and the ordinate the term $D(f(\mathbf{z}), \mathbf{y}^*)$ of equation (1). In the first place, $\text{RB-}p_{\text{Uni}}$ and $\text{RB-}p_{\text{OT}}$ perform markedly worse than the remainder, which attests to the importance of harnessing CE information when attributing features in pursuit of modification minimality in CE. In the second place, the observation that $\text{CF-}p_{\text{OT}}$ outperforms $\text{RB-}p_{\text{OT}}$ demonstrates that the deployment of OT augments the performance of p -SHAP specifically by furnishing an effective factual–counterfactual alignment, rather than by dint of any extraneous factor, since the two differ solely in A_{Shap} (see Table 2). In the third place, p -SHAP outperforms $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$ by a considerable margin, which bespeaks the efficacy of the joint distribution recovered by OT within p -SHAP: that is to say, the mere use of counterfactual information for FA (as practised by $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$) does not suffice, and a proper alignment—which need

not coincide with the one dictated by the exact counterfactual generation mechanism, as Figure 4 subsequently reveals—between factual and counterfactual must be taken into account.

We have observed that COLA equipped with p -SHAP exhibits salutary performance across many of the experiment scenarios defined in Table 1, as evidenced in Figure 3 as well as Table 3, and a copious body of further such results is adduced in Appendix H. In sum, p -SHAP outperforms every competing Shapley method in respect of sparsity in CE.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks This result attests to the efficacy of p -SHAP in expunging the influence of the CE generation process, effected by supplanting the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between the factual and counterfactual data, as displayed in Figure 4. We benchmark the method CF- p_{Ect} by means of COLA, directing our attention to the solution of equation (1) with MeanD as the divergence function D . Since CF- p_{Ect} reposes upon a known factual-counterfactual alignment, we thereby benchmark the effectiveness with which COLA exploits this alignment. The theoretical optimum of COLA in this case may be obtained by solving an MILP; see Appendix G for the derivation of the MILP formulation. As solving the MILP is computationally onerous, it is undertaken for *German Credit* alone.

One observes that CF- p_{Ect} attains a near-optimal performance under DiCE. It bears remarking that, for DiCE and KNN, the factual-counterfactual pairs are mutually independent, whence we contend that COLA is effective for instance-based CE, notwithstanding that our formulation in equation (1) is cast as a generalisation encompassing group or distributional CE. It is instructive to note that CF- p_{OT} on occasion outperforms CF- p_{Ect} , inasmuch as it invokes a more theoretically grounded procedure for identifying the key features warranting modification, whereas CF- p_{Ect} leans upon CE algorithm-dependent knowledge whose efficacy for FA wants solid justification. We remark that a gap persists between CF- p_{Ect} , CF- p_{OT} , and the optimal result under KNN; the determination of the best alignment remains an open question.

7 CONCLUSIONS

This paper has introduced a novel framework, COLA, for the refinement of CE by means of joint-distribution-informed Shapley values, whereby the refined CE is made to preserve the counterfactual effect while exacting fewer actions.

ETHICS STATEMENT

All authors have read and undertake to abide by the ICLR Code of Ethics.⁴ This paper advances a method (*COLA* with p -SHAP) for the refinement of CE. Our work being methodological and empirical in character, it neither implicates human subjects nor entails interventions, and we have neither collected, annotated, nor released any new personal data.

All experiments avail themselves of standard, de-identified, publicly available tabular datasets (for example, Adult/Census Income, German Credit, HELOC). We adhere to the licences and terms governing these datasets and to the prevailing research convention of employing such data solely for the purpose of benchmarking; we attempt neither re-identification nor linkage, and we release no additional personal information. Our code will, moreover, incorporate provisions for excluding protected attributes and for enforcing feasibility and immutability constraints upon features.

CE may bring to light the biases residing in the underlying models, yet they do not of themselves guarantee fairness. Our method can diminish the number of recommended feature edits; it does *not*, however, sanction edits to sensitive or immutable attributes (such as sex, race, or age), nor does it confer causal validity. We deprecate the use of CE to “game” high-stakes systems (in lending, hiring, or healthcare, say) or to coerce individuals into unrealistic or unethical alterations of behaviour. Any deployment within consequential domains ought to incorporate feasibility constraints defined

³This knowledge—namely the exact alignment between factual and counterfactual—is furnished, among all the CE methods under consideration, only by DiCE and KNN.

⁴<https://iclr.cc/public/CodeOfEthics>

by domain experts, fairness audits (for instance, disparate impact or benefit across groups), human oversight, and compliance with the applicable data-protection and AI regulations. We caution explicitly that our attribution is distributional rather than causal, and must on no account be construed as a causal effect.

The attendant risks encompass (i) the misconstrual of attributions as causal, (ii) the recommendation of infeasible or harmful edits, and (iii) the facilitation of strategic manipulation. We mitigate these by (a) stipulating the non-causal scope and limitations within the paper, (b) supporting cost and feasibility masks together with immutable features in the implementation, (c) reporting performance across subgroups where pertinent, and (d) furnishing guidance for responsible use in the documentation.

We provide an anonymised implementation, detailed settings, and seeds so as to facilitate replication; we report the model types, hyperparameters, and compute budgets, and we forestall the cherry-picking of results by adhering to fixed evaluation protocols. To our knowledge, no conflicts of interest or sensitive sponsorships attach to this submission; any funding disclosures will be provided subsequent to the double-blind review.

REPRODUCIBILITY STATEMENT

We regard reproducibility as a matter of consequence and accordingly furnish pointers to every requisite component. The COLA framework and the p-SHAP attribution are specified by means of pseudocode and notation in the main paper (Algorithm 1; the Method and Preliminaries sections), while all theoretical assumptions together with the complete proofs are deferred to the appendix (the theorems and lemmas accompanied by line-by-line proofs).

The implementation details—hyperparameters, optimisation settings, environment specifications, and random seeds—are documented in Appendix I, which additionally describes the datasets employed and the evaluation protocol (encompassing the data splits and preprocessing). The baselines (models and CE generators) and the coupling choices are enumerated within the experimental sections and ablations, their configurations being mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; we further report the hardware and runtime in Appendix I so as to contextualise the compute requirements, and we supply ablation and sensitivity studies together with the small-scale MILP benchmark setup in the appendix in order to underpin independent verification.

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APPENDICES

A	Comparison to Existing Approaches	15
B	$\mathcal{NP}$-hardness of the Problem in equation (1)	15
C	Proof of Theorem 4.1: p-SHAP Towards Counterfactual Effect	16
D	Proof of Theorem 4.2: Interventional Effect	18
E	Proof of Theorem 5.1: Counterfactual Proximity	19
F	Analysis of Computational Complexity of COLA	20
G	An MILP Formulation of equation (1) With MeanD	20
H	Extended Numerical Results	21
I	Experiments Reproducibility	22

A COMPARISON TO EXISTING APPROACHES

The authors of (Albini et al., 2022) propounded CF-SHAP, which enlists counterfactual data points as the background distribution for the Shapley computation. This proposal, however, presupposes that the counterfactual data distribution is defined conditionally upon each individual data instance, a stipulation that in turn implies a known probabilistic alignment between every factual and every counterfactual instance; kindred assumptions are entertained in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). In the numerous scenarios in which global explanations are sought, this assumption breaks down. We note that the counterfactual-data-distribution setup of (Albini et al., 2022) constitutes a special case of ours. The authors of (Kwon & Zou, 2022) put forward WeightedSHAP, which apportions weights among the features rather than treating them as equal contributors; our proposed method, by contrast, weights the contributions of the data points and admits of a straightforward extension to the weighting of both rows and columns. The work of (Kommiya Mothilal et al., 2021) erects a framework for harnessing both FA and CE in the service of explainability; yet its CE-based FA labours under the same assumption as (Albini et al., 2022), which renders its generalisation to the group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025) settings problematic. A further pertinent contribution is (Sharma et al.), a method that models counterfactuals by recourse to a system’s causal structure and time-series predictors so as to decompose, in an equitable manner, a single observed change in a system-level metric into the contributions of the individual inputs. More significantly still, none of the aforementioned literature addresses the minimal-actions CE problem that forms the locus of the present paper.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) lie in close proximity to the one investigated herein. In (Karimi et al., 2021) the authors minimise the cost of performing actions under the assumption of a known structural causal model (SCM), which is seldom available in practice. The authors of (Kanamori et al., 2022) observe that it remains an open question whether existing CE methods can be pressed into service for solving that problem, and propose an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, erected upon the works (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). The solution of an MILP is, however, costly, and hence resistant to scaling.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below asserts that no scalable exact algorithm for solving the problem in its full generality is to be anticipated, unless $\mathcal{P} = \mathcal{NP}$, the hardness being attributable not merely to the non-linearity of f but equally to its combinatorial character.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the

objective is to determine a vector $\mathbf{z} \in \mathbb{R}^n$ that minimises the residual error while comprising at most K non-zero elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements of $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard, owing to the combinatorial nature of selecting the subset of variables to be admitted into the model.

We shall map this SR problem onto our problem equation (1) by means of the following settings. Let the number of features be $d = 1$ and the number of instances be n (coinciding with the dimension of the SR problem); let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and let the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as prescribed by the SR problem). We define the model f as the linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

As regards the model output, we set $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, that is, the squared Euclidean distance; as regards the instances, we set $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, inasmuch as $\mathbf{x} = \mathbf{0}$. We fix the maximum number of admissible feature changes C to equal k (the sparsity level of the SR problem), and the large constant M may be taken to be any sufficiently large positive number—for instance, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) reduce to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

It follows that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, whence $z_i = 0$; whereas if $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) accordingly becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus assumes the form:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation the variables c_i signal whether the variable z_i is permitted to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce $z_i = 0$ whenever $c_i = 0$, thereby mirroring the sparsity constraint of the SR problem, while the constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features may change, in correspondence with the sparsity level; the objective function, moreover, is identical to that of SR.

Since our formulation replicates verbatim the SR problem with a cardinality constraint—which is known to be $\mathcal{NP}$ -hard—the solution of Problem equation (1) is at least as hard as the solution of the SR problem. Consequently, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are the standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below establishes that the feature attributions are aligned with the expected costs of feature modification, thereby giving rise to action plans that are cost-efficient in the realisation of counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , it holds for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$ that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying either side of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing either side over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

In order to bound E_d yet more tightly, we invoke the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$, inasmuch as $\mathbf{p}_{OT}$ constitutes a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that, for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimises the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ denote the joint probability of $\mathbf{x}$ and $\mathbf{r}$ furnished by A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are fixed at $\mathbf{x}_{i,S}$, whereas the features in $\mathcal{F} \setminus S$ persist in being distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. The expected value of $f(\mathbf{r})$ under the intervention is therefore:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of $v^{(i)}(S)$, we have

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene so as to set the features in S to $\mathbf{x}_{i,S}$, an intervention denoted $do(\mathbf{r}_S = \mathbf{x}_{i,S})$ and independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Consequently, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the conjoint effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ denote the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For those elements at which $c_{ik} = 1$ (the modified elements), $z_{ik} = q_{ik}$, whereas for those at which $c_{ik} = 0$, $z_{ik} = x_{ik}$. We may therefore write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms relative to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds by virtue of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ reduces to a permutation matrix associated with a bijection σ . In consequence, the value-assignment algorithm A_{value} (whether maximising or averaging) assigns $q_i = r_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ by the permuted matrix $\mathbf{r}_\sigma$ in the general bound established in Theorem 5.1 (namely $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) yields the result forthwith. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We undertake the analysis of the computational complexity of COLA as follows.

First, we analyse A_{Prob} . Should the alignment between $\mathbf{x}$ and $\mathbf{r}$ be known a priori, A_{Prob} merely constructs the matrix $\mathbf{p}$ from this prior knowledge, at a cost of $\mathcal{O}(n \times m)$; otherwise, we contemplate solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, incurs a cost of $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

We next analyse A_{Shap} . For each subset the model is evaluated on all n data points, whence $\mathcal{O}(n)$ evaluations per subset. The incorporation of baseline values drawn from the reference data $\mathbf{r}$ entails substituting the values of certain features by their corresponding baseline values; this operation is $\mathcal{O}(m)$, inasmuch as it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. On the supposition that the reference values may be precomputed and accessed in constant time, the complexity of incorporating them may be regarded as $\mathcal{O}(d)$. The tally of M_{Shap} subsets occasions M_{Shap} model evaluations. Aggregating the foregoing steps, the complexity of A_{Shap} amounts to $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalisation in line 4 of COLA incurs a cost of $\mathcal{O}(n \times d)$.

In order to juxtapose the complexities of the two algorithms $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, we analyse each in a step-by-step manner. For $A_{\text{Value}}^{\max}$, for each row $\mathbf{x}_i$ of the data table it is necessary to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which entails $\mathcal{O}(m)$ operations per row; (2) identify the row $\mathbf{r}_j$ of the reference data bearing the highest probability p_{ij} , which entails $\mathcal{O}(m)$ operations per row; and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which entails $\mathcal{O}(d)$ operations per row. Since the data table comprises n rows, the total complexity of $A_{\text{Value}}^{\max}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ of the data table it is necessary to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which entails $\mathcal{O}(m)$ operations per row; 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which entails $\mathcal{O}(m)$ operations per row; and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which entails $\mathcal{O}(m \times d)$ operations per row. Since the data table comprises n rows, the total complexity of $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$, by virtue of the fact that $C \leq n \times d$.

Consequently, the complexity of COLA employing $A_{\text{Value}}^{\max}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

while the complexity of COLA employing $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA as a function of n , m , d , and the regularisation parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ should $A_{\text{Value}}^{\max}$ be employed and $N = \mathcal{O}(nmd)$ should $A_{\text{Value}}^{\text{avg}}$ be employed.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section we furnish a global-optimality benchmark for the exploitation of a known alignment between factual and counterfactual in solving equation (1) with D taken to be MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is employed, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ remains closer to $\mathbf{x}$ than $\mathbf{r}$ does, whereupon equation (1b) may be dispensed with. The formulation of equation (1) accordingly becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) coalesce into equation (13c), since CF- p_{Ect} is stipulated to be used. That is to say, for any $\mathbf{x}_i$ there exists an exact $\mathbf{r}_j$ that serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either remains unchanged or may be changed to r_{jk} ; hence $z_{ik} = r_{jk}c_{ik} + x_{ik}(1 - c_{ik})$, whose value is contingent upon the binary variable c_{ik} .

Owing to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, the matrix $\mathbf{q}$ is determined (and, moreover, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the identical $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the solution $\mathbf{z}_i$ in which all features $k \in S$ are changed to q_{ik} while the remaining features $h \in \mathcal{F} \setminus S$ retain the value x_{ih} . The set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) thus constitutes the domain of all possible values of $\mathbf{z}$. We define a corresponding scalar variable for each $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ admits of representation by a binary variable a_{iS} in concert with the scalar g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimisation problem equation (13) is hence recast as the mixed integer program below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimising η subject to the two constraints equation (14b) and equation (14c) is equivalent to minimising the objective function of equation (13). The constraint equation (14d) guarantees that each data point i is subject to one and only one feature-modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$), while the constraint equation (14e) corresponds to equation (13b). The solution of equation (14) yields the theoretical optimum of COLA under a known alignment between factual and counterfactual, as demonstrated in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

On inspection of Figures 5–8, CF- p_{Uni} generally outperforms RB- p_{Uni} . Second, consideration of RB- p_{OT} and CF- p_{OT} —which likewise differ solely in A_{Shap} —reveals that the latter consistently surpasses the former. It follows that RB-SHAP is ill-suited to FA in CE.

We analyse the manner in which the several Shapley methods bear upon FA, corresponding to lines 3–4 of COLA. These methods fall into two categories. In the first place, consider $RB-p_{\text{Uni}}$ and $CF-p_{\text{Uni}}$, which differ only in A_{Shap} : on inspection of Figures 5–8, $CF-p_{\text{Uni}}$ generally outperforms $RB-p_{\text{Uni}}$. In the second place, consider $RB-p_{\text{OT}}$ and $CF-p_{\text{OT}}$, which also differ only in A_{Shap} : the latter consistently surpasses the former. Hence, once more, *RB-SHAP is ill-suited to FA in CE*.

Besides FA, a further and equally consequential step of COLA is line 5, namely the use of the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$, which apprises the factual x of the direction in which its features are to be changed so as to advance towards the target model outcome. We observe in Figures 5–8 that $CF-p_{\text{OT}}$ consistently outperforms every other method throughout all the experiments. It bears noting that all three methods $CF-p_{\text{Uni}}$, $CF-p_{\text{Rnd}}$, and $CF-p_{\text{OT}}$ supply solutions for the joint probability $\mathbf{p}$ when the exact alignment between factuals and counterfactuals is unknown; yet their performance diverges appreciably, whence the mere knowledge of the CE (and its marginal distribution) proves insufficient.

OT proves exceptionally serviceable whenever the alignment information between factual and counterfactual instances is absent or inaccurate. Even when the CE algorithm explicitly matches each factual instance to a corresponding counterfactual, it is difficult to warrant the claim that the known alignment optimises performance—a difficulty corroborated by Figure 4 of Section 6.

It is to be noted that p_{OT} need not be the true joint distribution of $\mathbf{x}$ and $\mathbf{r}$ from the vantage of data generation; rather, its office is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ conjointly for the purposes of both FA and CE. Furthermore, the QDA column of Figure 5 attests to the stability of the OT-based methods, whereas the others diverge markedly from the target. We emphasise, however, that COLA is *by no means confined to the use of OT* as A_{Prob} . As Figure 4 intimates, even the best known $\mathbf{p}$ leaves a non-negligible gap to global optimality; the design of a superior A_{Prob} algorithm is therefore an intriguing topic that merits further exploration.

I EXPERIMENTS REPRODUCIBILITY

The experiments are conducted upon a high-performance computing (HPC) cluster operating with four nodes (one per dataset) in parallel, each node being furnished with two Intel Xeon Processor 2660v3 units (10 cores, 2.60GHz) and 128 GB of memory. Each node runs for approximately 5–10 hours, contingent upon the size of the dataset. Reproduction upon a laptop is likewise feasible, albeit at the cost of a generally greater expenditure of computational time than the HPC cluster demands.

For all four datasets the numerical features are standardised, while the categorical features are subjected to either label encoding or one-hot encoding; in practice we discerned no appreciable difference between the two encoding schemes as regards the performance of COLA. The train–test split adheres to a ratio of 7 : 3.

The optimality baseline exhibited in Figure 4 is solved by means of Gurobi 11.0.2 ([gur](#)), whose reproduction presupposes a Gurobi licence; failing this, one may have recourse to open-source operations-research libraries such as the Google-OR tools ([goo](#)). We remark that solving the MILP of Appendix G is computationally expensive, so that it is applicable only to small-scale datasets such as German Credit; should one wish to compute the optimality baseline for other datasets, the number of features employed must be curtailed.

The hyperparameters of the models used in the experiments are specified as follows. The models Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)), with all hyperparameters retained at their default values. The models DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN comprises three layers and the SVM employs the linear kernel. The models XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are accessed through their scikit-learn interface, again with all hyperparameters retained at their defaults.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We availed ourselves of large language models (LLMs) solely as general-purpose assistive tools, their role being confined to grammar, wording, and light copy-editing of author-written text. LLMs contributed neither to the research ideation, nor to the modelling choices, nor to the experimental

1188 design, nor to the results. All algorithms, proofs, datasets, and analyses were devised and verified by
1189 the authors, and any phrasing suggested by an LLM was reviewed and revised prior to its inclusion in
1190 the paper. The authors assume full responsibility for the entirety of the content; LLMs are neither
1191 authors nor contributors. This disclosure conforms to the ICLR policy governing LLM usage.
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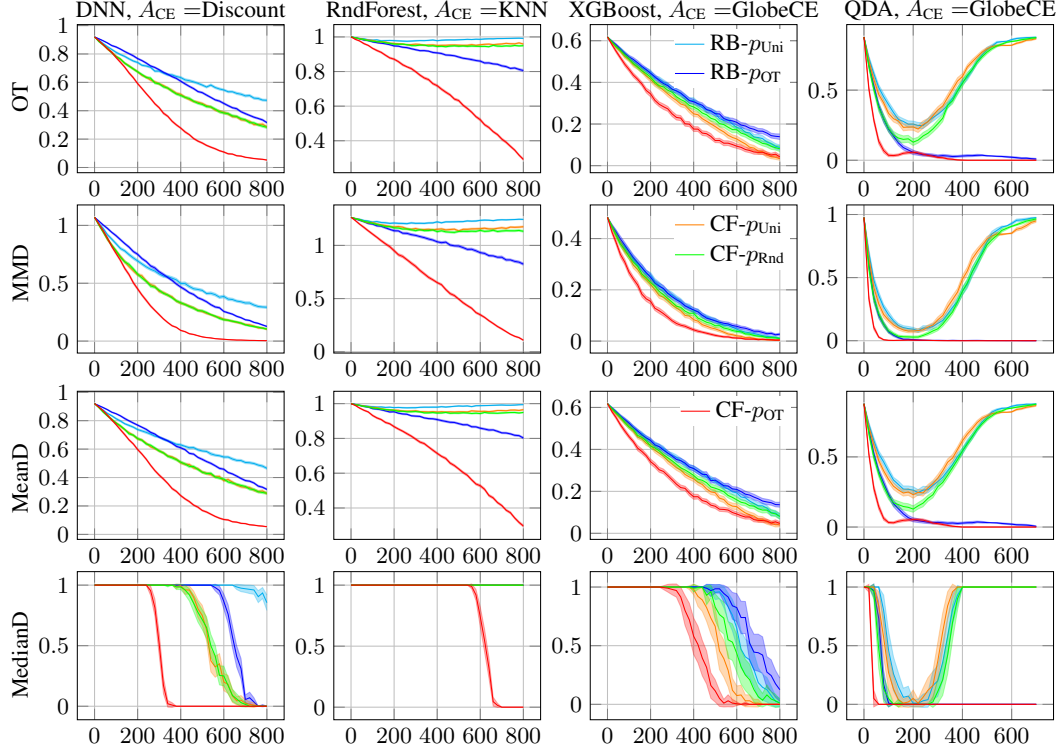


Figure 5: *[HELOC]* $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C . Each experiment comprises 100 runs, and the shaded regions delineate the 99.9% confidence intervals. The legends pertain to all plots. $A_{\text{Value}}^{\text{avg}}$ is employed.

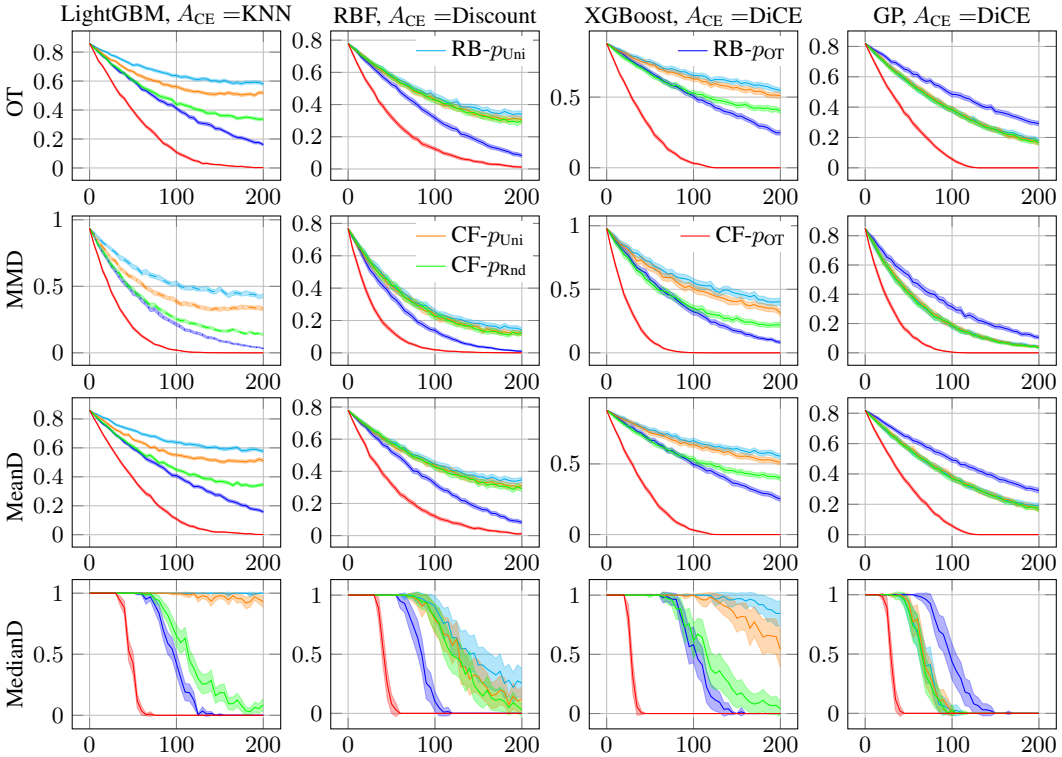


Figure 6: *[German Credit]* $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C . Each experiment comprises 100 runs, and the shaded regions delineate the 99.9% confidence intervals. The legends pertain to all plots. $A_{\text{Value}}^{\text{avg}}$ is employed.

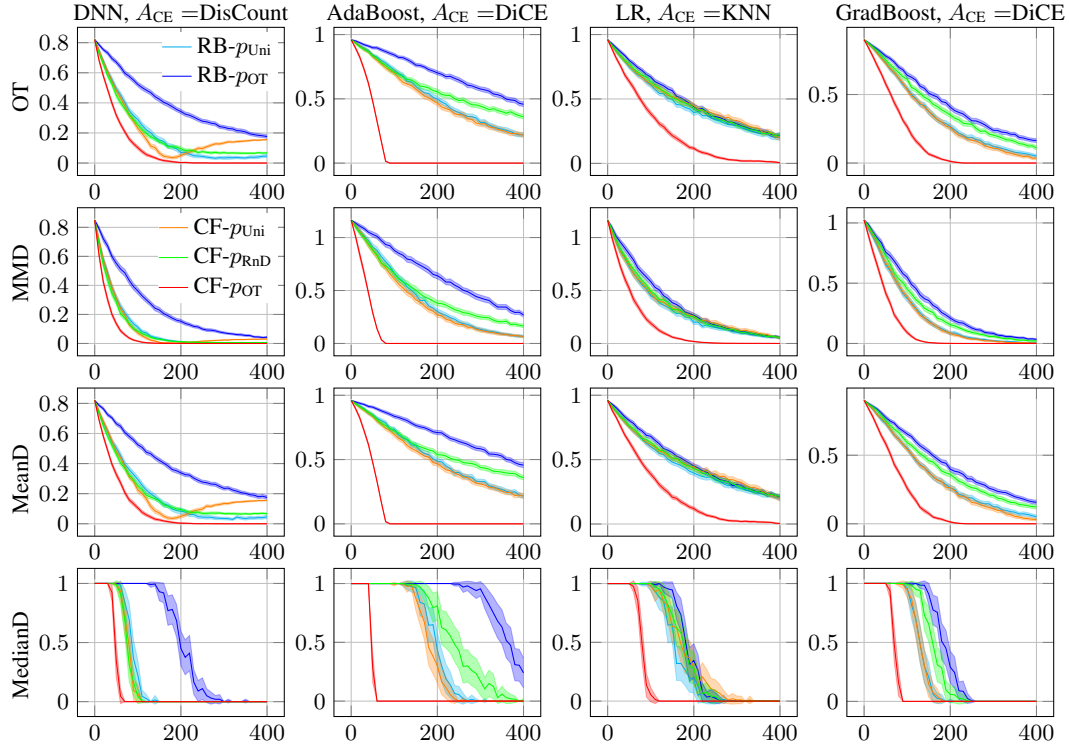


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C . Each experiment comprises 100 runs, and the shaded regions delineate the 99.9% confidence intervals. The legends pertain to all plots. $A_{\text{Value}}^{\max}$ is employed.

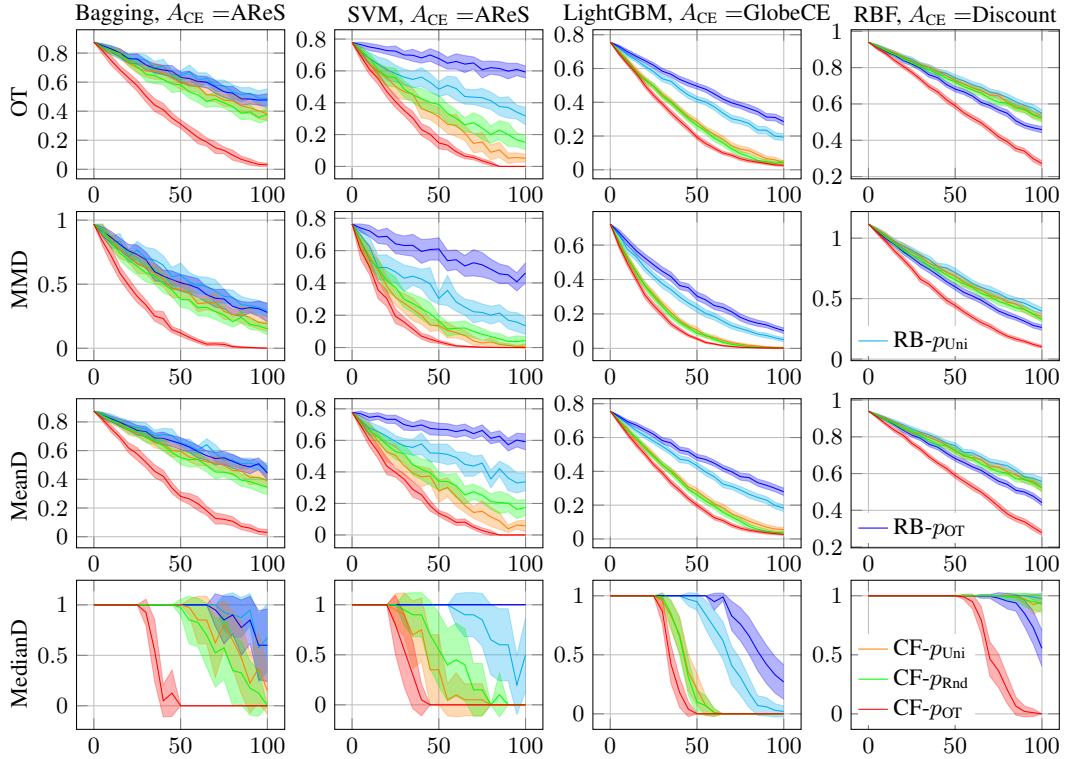


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ plotted against the number of allowed actions C . Each experiment comprises 100 runs, and the shaded regions delineate the 99.9% confidence intervals. The legends pertain to all plots. $A_{\text{Value}}^{\max}$ is employed.